

THINKING IN SYSTEMS

A Primer

DONELLA H. MEADOWS



The Big “So What”

We live in a complex world of inter-locking systems. Systems thinking helps us to understand how things work, so we can better identify root problem causes, see new opportunities, make better decisions and adapt to changing circumstances. In this summary, we will:

- Give an overview of systems thinking concepts and principles; and
- Explain how to apply systems thinking in real-life.



Systems: Concepts and Principles



What are Systems?



Definition: A system is a set of interlinked elements that are organized to achieve a goal.

- Every animal, plant, organization and society is a complex system. A system's structure defines its behavioral pattern. While systems may be influenced by external forces, the way they respond to these forces tend to come from their inherent characteristics.
- This means that the economy will move in cycles regardless of what political leaders do. Companies that lose market share would've done so regardless of how competitors respond. Unless we can see how systems operate and create their own problems, we can't identify and address the root causes. We'll be like the proverbial blind men, each touching a different part of an elephant and drawing different conclusions about what an elephant is like.

KEY QUOTES

“We know a tremendous amount about how the world works, but not nearly enough.”

“Systems happen all at once. They are interconnected not just in one direction, but in many directions simultaneously.”



A system is more than the sum of its parts. Its elements are interrelated, e.g. the organs in your digestive system jointly break down the food you consume. Systems can be a part of larger systems, and may be able to achieve goals, evolve or even self-repair.

- System elements can be physical (e.g. machines, people, products) or intangible (e.g. culture, values, skillsets).
- Due to the sheer number of elements and sub-elements, it's more useful to examine the interconnections between elements—these may be physical (e.g. how products move through a factory) or information-based (e.g. how a customer decides to buy something).
- A system's purpose is the hardest to decipher—the only way to do so is to observe exactly how the system behaves. Since each sub-system has its own goals, a system can have a complex network of sub-systems and goals. For example, the students and staff in a university will have different goals from the university itself.
- Elements are the most observable but least crucial. The greatest impact comes from changes to the system's purpose, followed by its interconnections and elements.



Stocks, Flows and Feedback Loops



Stocks and flows help us to understand how a system behaves.

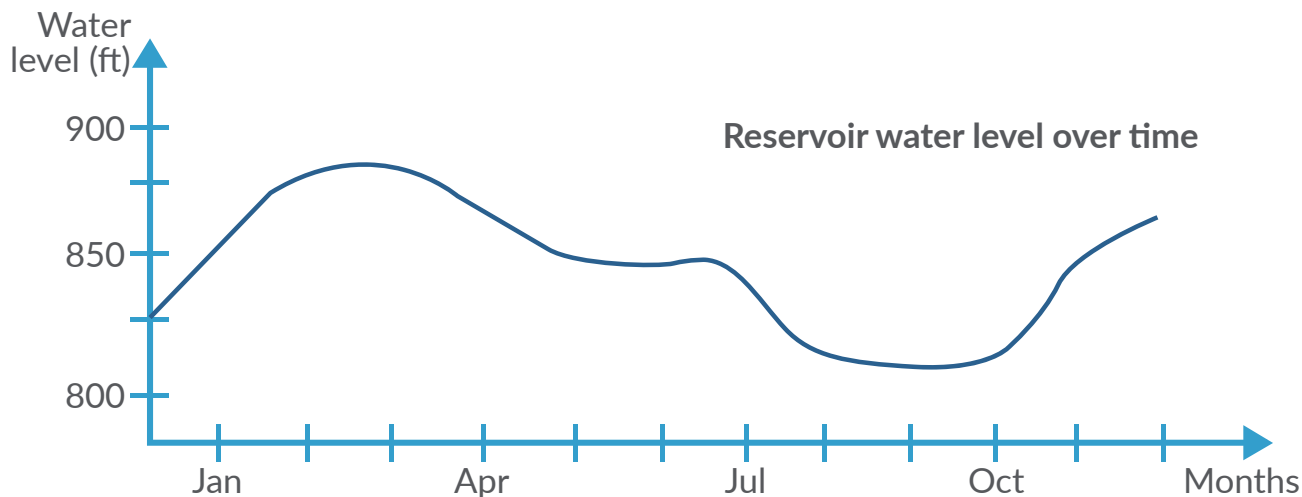
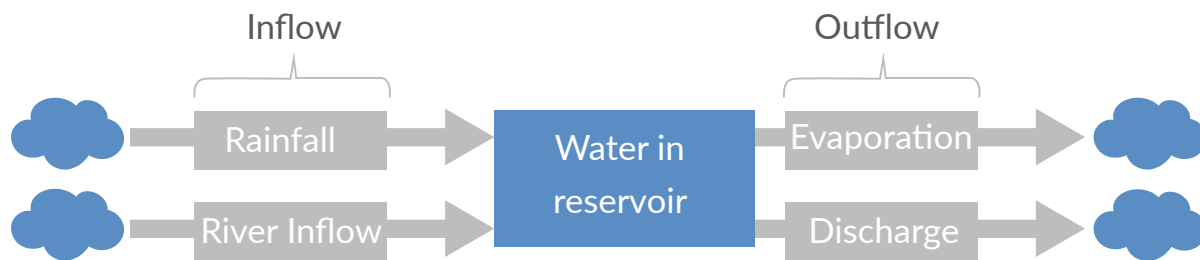
- A stock is an observable or quantifiable aspect of a system, e.g. population size or inventory levels. It can be accumulated or depleted, and thus captures the history of inflows and outflows in the system. Stock levels rise when inflow is greater than outflow and vice versa. Stock levels stay constant and achieve dynamic equilibrium when inflow = outflow.
- The best way to understand a system is to (i) map out a system diagram and (ii) plot the variables/stocks on a time graph. For example, the simple diagram below shows the key factors that affect the water level in a reservoir. When you examine it alongside time-graphs, you can get a better grasp of how each variable affects the water levels.

KEY QUOTES

"The behavior of a system cannot be known just by knowing the elements of which the system is made."

"Purposes are deduced from behavior, not from rhetoric or stated goals."

"A change in purpose changes a system profoundly, even if every element and interconnection remains the same."



- Stock levels tend to change gradually, thus providing stability or momentum in a system. We use stock levels to make decisions (e.g. offer discounts when there's excess inventory, spend less when the bank balance is low). Remember: flows take time and things can only happen as fast as they happen. To achieve your desired stock levels, you need to take such delays into account.



A feedback loop is formed when a change in stock affects the inflow/outflow for that stock, i.e. it triggers a process to increase, decrease or maintain the stock level. When you see a consistent pattern of behavior, there's always a feedback loop at work.

- A balancing loop is goal-seeking. For example, a thermostat regulates temperature—it warms up a room if it gets too cold, and cools it down if it gets too hot. Such loops bring stability but also resist change.
- A reinforcing loop amplifies an existing effect to trigger a virtuous or vicious cycle. For example, the more people catch a virus the faster it spreads, or the more you save the faster your bank balance grows from the compound interest. Such loops bring exponential growth but can also cause things to spin out of control and collapse.



Systems Behaviors



Just like how animals in the zoo give us a snapshot (but not the full picture) of wildlife, systems diagrams represent a simplified

KEY QUOTES

"A stock is the memory of the history of changing flows within the system."

"A stock takes time to change, because flows take time to flow."

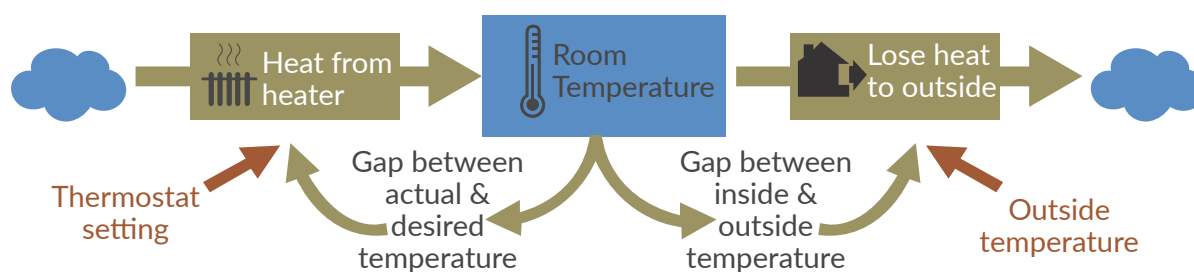
"Systems with similar feedback structures produce similar dynamic behaviors."

model of the world. Two systems that appear very different can exhibit similar patterns so long as they have similar feedback loops.



A one-stock system can have many feedback loops.

- During winter, the room temperature is affected by the thermostat setting (which controls heat released from the heater) and the outside temperature (which affects heat loss). These are 2 conflicting balancing loops that affect the room temperature. A feedback loop can break down (e.g. if all the windows are opened to cause excessive heat loss).



- A country's population is determined by the nett effect of a reinforcing loop (# births) and a balancing loop (# deaths). These are in turn driven by fertility and mortality rates. It's always easier to predict behavioral patterns than actual numbers, e.g. the exact population size could fluctuate as (i) fertility/mortality rates change over time, (ii) a feedback loop becomes more dominant, and/or (iii) additional factors emerge (e.g. migration).



- There's always a time lag between something happening and the feedback loop registering it and taking corrective action. In balancing loops, a delay in 1 part of the system is likely to lead to oscillation (e.g. delay in product delivery → shop runs out of stock → perceived shortfall → place larger order → overstock). This effect, when combined with imperfect information and multiple feedback loops operating concurrently, can create massive ripples throughout a system.



A two-stock system can be much more complex, especially when 1 stock limits another stock, e.g. a mining company's money being limited by the availability of precious metals in the mines.

KEY QUOTES

"Dynamic systems studies usually are not designed to predict what will happen. Rather, they're designed to explore what would happen if a number of driving factors unfold in a range of different ways."

"We can't begin to understand the dynamic behavior of systems unless we know where and how long the delays are."

"Any real physical entity is always surrounded by and exchanging things with its environment."

- No physical entity/system can grow indefinitely—at some point, balancing loops will kick in to limit growth, e.g. a company can only grow insofar as it has manpower, customers, materials and resources. The nature of the constraints depend on whether the critical resources for growth are renewable or non-renewable.
- Non-renewable stocks are fixed in supply, e.g. once an oil field is depleted, there'll be no more oil. Your rate of extraction is only limited by your capital or resources. However, there's an inherent balancing loop in place: the faster you grow, the faster you'll drain the resources, making it progressively harder to grow until the stock is totally depleted.
- Renewable stocks can be living (e.g. fish or grass) or non-living (e.g. sunlight or wind), and can be regenerated through a reinforcing loop or a constant supply. They're limited by flow, i.e. they're inexhaustible so long as they're harvested at a rate slower than their rate of regeneration. However, once they're driven below a critical threshold, they can become non-renewable. For example, in overpopulated waters, fishes' rate of reproduction may improve initially if a slight decrease in fishes leaves more food for the remaining fishes. However, once the population drops beyond a critical point, reproduction plummets and the fish can become a non-renewable resource.

KEY QUOTES

"A quantity growing exponentially toward a constraint or limit reaches that limit in a surprisingly short time."

"Nonrenewable resources are stock-limited... Renewable resources are flow-limited."



Applying Systems Thinking



3 Characteristics of Effective Systems

Systems thrive with 3 key characteristics: resilience, self-organization and hierarchy.

→||← Resilience in this book refers to elasticity, or the ability to bounce back to the original state after being strained or manipulated.

- Resilient systems are dynamic. A system is more resilient when (i) it has multiple feedback loops (so when 1 loop fails, another kicks in) and (ii) there's at least 1 loop that can repair other loops. The most resilient systems are those that can self-organize.
- Unfortunately, resilience is not obvious unless you have a full system perspective. We tend to sacrifice resilience for stability or productivity. For example, growth hormones can increase a

"Systems need to be managed not only for productivity or stability, they also need to be managed for resilience—the ability to recover from perturbation, the ability to restore or repair themselves."

cow's milk-production but reduces its long-term immunity. Pesticides kill pests, but they also kill off other species to disturb the eco-system's bio-chemistry, reduce its resilience and trigger long-term ecological disasters.

- Look beyond productivity or observable properties to consider how you can strengthen a system's resilience and self-regeneration capabilities. For example, instead of simply giving food and money to the needy, develop programs that increase people's ability to generate their own food and money.



Self-organization is the ability to learn, evolve and create new complex systems/structures to survive, rebuild and thrive.

- For example, the human body is developed from a single cell and constantly repairs itself. A company converts raw materials into a much more sophisticated form (e.g. a car or computer).
- Self-organization is spontaneous and unpredictable; it requires freedom and experimentation. However, complex creations are often built upon a few simple rules, like how all lifeforms are captured in a series of DNA, RNA and protein molecules.



Complex systems require hierarchies to organize the huge amounts of information/interactions and operate stably.

- For example, each cell in our body is part of an organ, which is a part of a system (e.g. digestive or reproductive system), which is a part of our human body. Each of us in turn is a part of a family, community, nation etc. Ideally, each sub-system should be able to manage itself while serving the needs of a larger system. The larger system in turn should integrate/augment the sub-systems synergistically.
- A system is sub-optimal when hierarchies don't work in sync. In such cases, imposing centralized control won't work and may even backfire. The only sustainable approach is to understand and address the needs of the sub-systems so the entire system can flourish dynamically. When hierarchies malfunction, the entire system can break down, e.g. when certain cells in the body multiply out of control and attack other cells.

KEY QUOTES

"Productivity and stability are the usual excuses for turning creative human beings into mechanical adjuncts to production processes."

"The ability to self-organize is the strongest form of system resilience. A system that can evolve can survive almost any change, by changing itself."

"Hierarchical systems evolve from the bottom up. The purpose of the upper layers of the hierarchy is to serve the purposes of the lower levels."

"When a subsystem's goals dominate at the expense of the total system's goals, the resulting behavior is called suboptimization."



Thinking in Systems



We tend to get surprised by systems because we over-estimate how much we know and under-estimate how complex systems are.

- Real-world systems are complex networks of stocks, flows, and feedback loops. Systems thinking helps us to consider the relationship between system structures, behaviors, and events. By asking what-ifs, we can consider possible scenarios so we're less likely to be surprised.
- Yet, in most cases, it's impossible to accurately predict the future. For example, there's no way to account for all the variables, relationships and events that affect economic growth or stock market indices. That's why complex modeling may still work for short-term predictions, but not for long-term forecasts.



Watch out for these thinking gaps:

- Linear vs nonlinear relationships. We tend to think in linear relationships (i.e. assume things run in a straight line) when most relationships are actually non-linear (i.e. there's no proportionate relationship between variables). For example, once vehicle density reaches a certain level, traffic jams get much worse.
- Imaginary boundaries. For simplicity, system diagrams use clouds to denote the stocks at the start/end. In reality, there's no finite start or end to a system. For instance, the earlier diagram on population size doesn't address the drivers for mortality/death rates, nor other factors that affect population size e.g. migration. It's unrealistic to try and capture *everything*—the key is to be aware that all system boundaries are imaginary.
- Multiple constraints. Every outcome is influenced by multiple factors, but we tend to think in terms of singular cause-effect relationships. Over time, a system's limiting factors (what's most vital for growth) will also change, e.g. a startup faces different constraints as it grows to a mid-sized firm. At some point, you must choose your limits (e.g. the maximum population size to maintain acceptable living standards), or the system will decide it for you (e.g. resource shortages, declining air quality).

KEY QUOTES

"Structure is the key to understanding not just what is happening, but why."

"The world is nonlinear. Trying to make it linear for our mathematical or administrative convenience is not usually a good idea."

"Remember that boundaries are of our own making, and that they can and should be reconsidered for each new discussion, problem, or purpose."

"Ultimately, the choice is not to grow forever but to decide what limits to live within."

- Pervasive delays. Things take time to happen, and there're delays in most feedback loops—delays in shipping, perception, processing etc. Mail takes time to reach the recipient and viruses take time to develop into diseases with observable symptoms. If there're long delays in the feedback loop, it'll be too late to take action when the symptoms surface, e.g. if you only start to build power plants when there're electricity shortages, you'll end up with cycles of under-capacity and over-capacity.
- Bounded rationality. We make the best possible decisions based on what we know to satisfy a current need. However, we (i) won't have perfect information, (ii) can't see all options, viewpoints or implications, and (iii) will always interpret info subjectively. Thus, a sound decision that meets a personal short-term goal can create long-term problems for ourselves and others. The best way to get a bigger picture is to experience the system from a different position, e.g. an activist going to work for the government body he's been campaigning against. Each shift gives you a wider system perspective to make better decisions.



Identifying System Traps and Opportunities

Learn to recognize the “behavior archetypes” or system structures behind common problems, so you can deliberately restructure them to fix the problems at cause.



Resistance to change. Balancing feedback loops can sustain behavioral patterns to make them change-resistant.

- In 1967, the Romanian government tried to increase birth rates by making abortions illegal for women under 45 years old. There was a temporary spike in birth rate, then people turned to illegal abortions or abandoned the children they couldn't afford to raise. So long as the sub-systems have different goals with players pulling in different directions, there'll be policy resistance. The harder 1 player pushes an outcome, the harder the other players in the balancing loop will resist.
- A better approach is to step back, identify all the players' goals and how to incentivize them to move in the desired direction. For example, to increase birth rates, you need to know why people aren't having more children. In the 1930s, Sweden's birth rate was declining. Instead of trying to forcibly increase family sizes,

KEY QUOTES

“To act only when a problem becomes obvious is to miss an important opportunity to solve the problem.”

“Change comes first from stepping outside the limited information that can be seen from any single place in the system and getting an overview.”

“The most effective way of dealing with policy resistance is to find a way of aligning the various goals of the subsystems, usually by providing an overarching goal that allows all actors to break out of their bounded rationality.”

the Swedish government rallied the country behind the common goal of improving the quality of child care so every child is wanted and nurtured.



The “tragedy of the commons” comes about when a shared stock is eroded or destroyed because people exploit more and more of the stock instead of conserving it.

- Such problems arise in any common resource (e.g. public parks, common toilets) where users benefit directly from the stock, but share the cost of stock depletion. For example, a farmer can directly use every tree he cuts down, whereas the costs of deforestation is shared with everyone else. The more users there are, the scarcer the resource and the more people rush to consume it before it's depleted.
- Solutions include: (i) educating people on the implications of uncontrolled usage and appeal to their good sense, (ii) privatizing/splitting the resource so people are accountable for their own usage and (iii) using regulations (e.g. taxes, quotas, permits) to limit consumption. Paid parking lots, traffic lights, broadcasting licenses etc. are all examples of ways to manage the commons.



Systems may experience sliding standards or a “drift to low performance” if (i) the players perceive things to be worse than they are, and (ii) such perceptions affect their future targets.

- For example, you may accept lower-than-desired air quality because it doesn't seem any worse than before. The new (lower) standard becomes the norm and the negative loop continues.
- Solutions include: (i) sticking to absolute goals (vs being influenced by past/existing performance) or (ii) benchmarking only against the best historical performance.



Escalation arises when competing players try to outdo one another in a reinforcing loop, e.g. price war, shouting match, weapons race.

- Even escalation for positive things (e.g. efficiency, morality) can become negative when taken to the extreme. If left unchecked, it usually ends up with at least 1 of the players collapsing.

KEY QUOTES

“The tragedy of the commons arises from missing (or too long delayed) feedback from the resource to the growth of the users of that resource.”

“Standards aren't absolute. When perceived performance slips, the goal is allowed to slip.”

- Solutions include: (i) one party taking the first steps to unilaterally reduce the stocks (and stop the reinforcing loop) or (ii) negotiating balancing loops that can keep escalations in check.



Widening winner-loser gaps occur when winners get rewarded with resources that help them to win even bigger in future. This creates a reinforcing loop that could eliminate losers from the system, e.g. the rich using their wealth and access to gain even more wealth and access.

- Solutions include: (i) Diversification (e.g. a company moving to a new niche to exploit new resources), (ii) adding balancing loops to prevent domination (e.g. antitrust laws) or equalize the playing field (e.g. handicaps for weaker players), and (iii) finding ways to break the reinforcing loop (e.g. progressive taxes, public welfare).



Addictions. Just like how people can get addicted to alcohol or drugs, players in a system can get addicted to a solution that eases problem symptoms without addressing the root causes.

- Such interventions are usually introduced to boost a stock. However, if the root cause isn't addressed, problems will resurface and demand an even bigger fix. Worse still, the intervention could cause a degeneration of the original system capabilities, and shift the burden increasingly to the intervener. For example, kids' reliance on calculators can reduce their capacity for mental arithmetic, and the use of pesticides can destroy the pests' natural predators so farmers need to use even more pesticides.
- Solutions include: (i) withdrawing from the addiction (gradually or going cold-turkey), or (ii) avoiding it from the onset by strengthening the system's abilities instead of taking on its burdens.



In any system, there'll be people trying to break the rules, i.e. they seem to be following the rules but are actually working around them, e.g. drivers keeping to the speed limit only when there're speed cameras or police cars. Use rule-beating as a form of feedback to refine the rules, or design rules that specifically incentivize people to self-organize in the desired way.

KEY QUOTES

"Addiction is finding a quick and dirty solution to the symptom of the problem, which prevents or distracts one from the harder and longerterm task of solving the real problem."

"The secret is to begin not with a heroic takeover, but with a series of questions."

"Wherever there are rules, there is likely to be rule beating."

"Systems...have a terrible tendency to produce exactly and only what you ask them to produce. Be careful what you ask them to produce."

 Systems will deliver whatever they're designed to deliver—if you set the wrong goals, you'll get disastrous outcomes. For example, an education system will deliver vastly different outcomes depending on whether you measure “good education” using expenditure or test results. Don't confuse efforts with results.

Creating Effective System Changes

 The best way to change a system is to (i) identify the leverage points where a tiny shift could trigger a large change in behavior, then (ii) figure out the right direction to push. This is a difficult process that requires hard work and an open mind (to explore new ideas and options).

 Use these leverage points in order of impact:

- Every system or society comes with its own paradigm, i.e. a set of unspoken assumptions and beliefs (about what's good, right, fair, etc.). Paradigm shifts can be achieved in any system with the right reinforcements and change agents. However, the ultimate leverage comes from *transcending* paradigms, i.e. realizing that there's no single “right” or “true” paradigm.
- A system's goals define its behaviors. Most systems (including cancer cells) seek to survive and grow. The key is to deliberately choose how far to grow and to what end, so we prevent 1 system's unchecked growth from destroying others.
- Increase a system's self-organization abilities so it can change dynamically. The best way to do so is to promote learning and evolution through diversity and experimentation.
- Use rules—including rewards, penalties and constraints—to directly shape behavior.
- Typically, system players will organize themselves with the right info. Design information flows to incentivize players to act in the desired way, e.g. show the increasing cost per unit to discourage people from consuming too much of a limited resource.
- Build in checks and balances to curb the momentum of reinforcing feedback loops to prevent long-term system collapse.

KEY QUOTES

“There are no cheap tickets to mastery. You have to work hard at it.”

“If no paradigm is right, you can choose whatever one will help to achieve your purpose. If you have no idea where to get a purpose, you can listen to the universe.”

“Missing information flows is one of the most common causes of system malfunction.”

- A complex system comes with many balancing feedback loops, each designed to maintain a sub-goal and jointly define the overall system strength. Ensure that each balancing loop is strong enough relative to the impact it's designed to achieve, e.g. governments can only manage big industry players if they have enough power. Be careful when removing balancing loops e.g. an emergency cooling system may seem unimportant since it's usually inactive, yet its removal will weaken the system.
- Adjust the length of delays in feedback loops (where possible) to influence system behaviors and the level of oscillations.
- A system's behaviors are determined by the structure of stocks, flows and their interconnection. Your best leverage is to design a great system from the onset, since it's hard to change a system once it's built. You can only improve existing systems by addressing the bottlenecks and limiting factors.
- Build stocks that are big relative to the flows, so they can act as a buffer to stabilize the system, e.g. a lake is much less likely to overflow than a bathtub.
- Numbers seldom result in real change (e.g. tweaking minimum wages or tax rates won't restructure the economy), unless the change is big enough to trigger other leverage points above.



Conclusion

Ultimately, it's impossible to fully understand dynamic, self-organizing, nonlinear systems, much less predict or control them. The real value of systems thinking is in helping us to understand the complexity of the world we live in, so we can manage our expectations, learn from surprises, and actively design/refine systems and find creative ways to bring our visions to life.

The book is packed with examples and diagrams (ranging from business to sociology and economics) to help us understand systems at work. Meadows also presents several rules of thumb to help us navigate more effectively in a world of systems. For more details, please visit academyforchange.org.

KEY QUOTES

"The only way to fix a system that is laid out poorly is to rebuild it, if you can."

"The future can't be predicted, but it can be envisioned and brought lovingly into being."

"The thing to do, when you don't know, is not to bluff and not to freeze, but to learn."